

5.4.4 Absolute vs. Incremental (XYZ vs. UVW)

Absolute (XYZ) and incremental positioning (UVW) define how the control interprets axis motion commands.

When you command axis motion using x , y , or z , the axes move to that position relative to the origin of the coordinate system currently in use.

When you command axis motion using $U(X)$, $V(Y)$, or $W(Z)$, the axes move to that position relative to the current position.

Absolute programming is useful in most situations. Incremental programming is more efficient for repetitive, equally spaced cuts.

5.5 Miscellaneous Codes

This section lists frequently used M-codes. Most programs have at least one M-code from each of the following families.

Refer to the M-code section of this manual, starting on page **343**, for a list of all M-codes with descriptions.



NOTE:

You can use only one M-code on each line of the program.

5.5.1 Tool Functions

The `Tnn00` code selects the next tool (nn) and offset (00).

FANUC Coordinate System

T-codes have the format `Txxyy` where xx specifies the tool number from 1 to the maximum number of stations on the turret; and yy specifies the tool geometry and tool wear indices from 1 to 50. The tool geometry x and z values are added to the work offsets. If tool nose compensation is used, yy specifies the tool geometry index for radius, taper, and tip. If $yy = 00$ no tool geometry or wear is applied.

Tool Offsets Applied by FANUC

Setting a negative tool wear in the tool wear offsets moves the tool further in the negative direction of the axis. Thus, for O.D. turning and facing, setting a negative offset in the X-axis results in a smaller diameter part and setting a negative value in the Z-axis results in more material being taken off the face.



NOTE:

There is no X or Z motion required prior to performing a tool change and it wastes time in most cases to return X or Z to the home position. However, you must position X or Z to a safe location prior to a tool change in order to prevent a crash between the tools and the fixture or part.

Low air pressure or insufficient volume reduces the pressure applied to the turret clamp/unclamp piston and slows down the turret index time or does not unclamp the turret.

To load or change tools:

1. Press **[POWER UP/RESTART]** or **[ZERO RETURN]** and then **[ALL]**.
The control moves the tool turret to a normal position.
2. Press **[MDI/DNC]** to toggle to MDI mode.
3. Press **[TURRET FWD]** or **[TURRET REV]**.
The machine indexes the turret to the next tool position.
Shows the current tool in the **Active Tool** window in the lower right of the display.
4. Press **[CURRENT COMMANDS]**.
Shows the current tool in the **Active Tool** display in the upper right of the screen.

5.5.2 Spindle Commands

There are (3) primary spindle M-code commands:

- M03 commands the spindle to turn in the forward direction.
- M04 commands the spindle to turn in the reverse direction.



NOTE:

*You can command the spindle speed with an **Snnnn** address code, where **nnnn** specifies the speed in rpm, but overrides from G50, G96, or G97 may apply to the actual spindle speed.*

- M05 commands the spindle to stop.